

Agentic AI Chapter 1: LLM Foundations, Tokenization & Inference Mechanics

Core Learning Objective: Understand the internal mechanics of Large Language Models (LLMs) from the transformer architecture up. Learn how Byte-Pair Encoding (BPE) tokenizes text, how autoregressive generation works, why the KV-Cache is critical to inference speed, and how sampling parameters (temperature, top-p) shape model outputs.

0. Zero-Prerequisite Foundations: What is an LLM Really Doing?

The "Super-Autocomplete & Scrabble Tile" Metaphor What is an LLM? Is it a conscious digital brain thinking complex thoughts?

Not at all. Think about typing a text message on your smartphone. When you type: `Merry Christmas and a happy new...` Your phone keyboard displays a tiny box suggesting the next word: `year`.

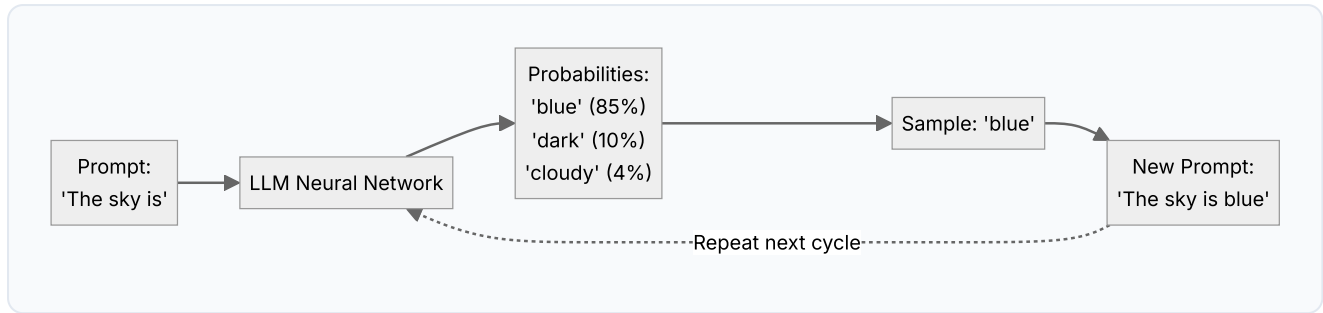
How did your phone know that? It didn't "think" about holidays or feel festive. It simply observed that across millions of past messages, after the words *"happy new"*, the word *"year"* appears 99% of the time.

A Large Language Model (like GPT-4o, Claude 3.5, or Llama 3) is fundamentally **that smartphone predictive keyboard scaled up to supercomputer proportions**. It was trained on trillions of words from books, GitHub codebases, scientific papers, and the internet.

How It Generates an Entire Response

When you ask ChatGPT: *"Write a Python function to reverse a string"*:

1. It reads your prompt.
2. It calculates the statistical probability for every possible next fragment of text in its vocabulary.
3. It picks the most probable fragment: `def`.
4. It appends `def` to the end of the text, and feeds the entire expanded text back into itself.
5. It predicts the next word: `reverse_string`.
6. It repeats this loop hundreds of times—like an ultra-fast typewriter hammering out one word fragment at a time! This sequential, one-word-at-a-time loop is called **Autoregressive Generation**.



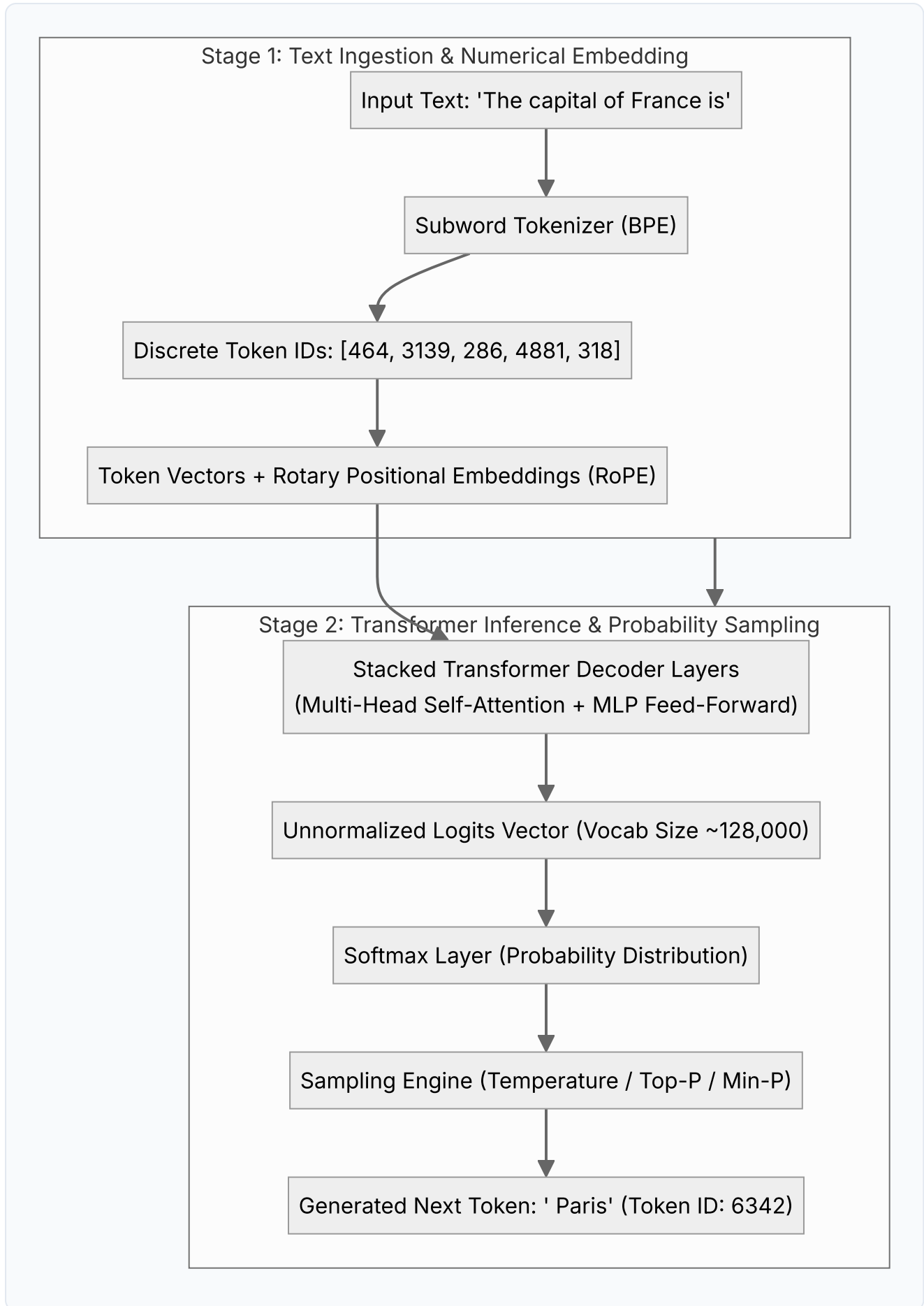
Core AI Jargon Demystified in Plain English:

- **Token:** Silicon chips cannot read English letters. They only understand numbers. A tokenizer is like a giant lookup dictionary that chops words into common letter clusters (tokens) and assigns each cluster a unique ID number. (e.g. The word "Apple" is Token ID 19253).
- **Logits:** At the end of its neural calculations, the model generates raw score numbers (votes) for all 128,000 tokens in its dictionary. These raw unnormalized scores are called **Logits**.
- **Softmax:** A mathematical formula that converts those raw logit scores into clean percentages (probabilities) that add up to 100%.
- **Temperature (T):** The "adventurousness" or randomness slider:
 - $T = 0.0$ (**Deterministic / Greedy**): Always picks the #1 highest-probability token. 100% repeatable. Best for code generation, SQL queries, and JSON extraction.
 - $T = 0.7 - 1.0$ (**Creative**): Takes calculated risks by occasionally picking the #2 or #3 option. Great for brainstorming, storytelling, and conversational variety.

1. The Core Nature of Large Language Models

An LLM is fundamentally a **statistical next-token prediction engine** trained on vast text corpora. It computes a probability distribution over a finite vocabulary of discrete numerical tokens:

$$P(w_t \mid w_1, w_2, \dots, w_{t-1})$$



2. Tokenization: Byte-Pair Encoding (BPE)

LLMs cannot process raw text strings directly. They rely on subword tokenizers like **Byte-Pair Encoding (BPE)**:

- A fixed vocabulary of subwords (e.g. GPT-4 has $\sim 100,000$ tokens, Llama 3 has 128,256 tokens).
- Common words ("the", "system") are a single token.
- Rare words, code, or non-English characters are broken into multiple subword pieces or raw bytes.

Inspecting Tokens with `tiktoken` in Python:

WARNING

Token Arithmetic Pitfall: LLMs struggle with basic character counts (e.g. *"How many 'r's are in strawberry?"*) because the model never sees individual letters—it only sees the token IDs representing subword chunks!

3. Autoregressive Inference & The KV-Cache

LLM text generation is **autoregressive**: to generate a 100-word response, the model must run 100 sequential forward passes through all transformer layers, predicting one single token at a time.



Why Inference is Memory-Bandwidth Bound

During the generation phase (decoding), the GPU does not bottleneck on mathematical compute FLOPS—it bottlenecks on **GPU VRAM memory bandwidth**, because it must reload gigabytes of cached Key and Value vectors from memory for every single generated token.

4. Sampling Hyperparameters: Shaping the Probability Distribution

After the final transformer layer, the model outputs raw unnormalized scores called **Logits** (z_i) for every token in the vocabulary. The **Softmax function** converts logits into probabilities:

$$P(i) = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

Where T is the **Temperature**:

Temperature = 0.0 (Greedy Decoding)
Sharpens distribution: unconditionally selects the highest probability token.
100% Deterministic. Ideal for Code Generation, Structured JSON Extraction & Math.

Temperature = 1.0 (Creative Sampling)
Flattens distribution: lower probability candidate tokens receive realistic selection odds.
High Diversity & Hallucination Risk. Ideal for Creative Ideation & Brainstorming.

Key Generation Parameters:

1. **Temperature (T)**: Scales logit variance. $T = 0$ yields greedy deterministic output.
2. **Top-P (Nucleus Sampling)**: Accumulates candidate tokens in descending order of probability until their cumulative probability reaches P (e.g., $P = 0.90$). Truncates the long tail of unlikely words.
3. **Top-K**: Restricts candidate selection to strictly the K highest probability tokens (e.g., $K = 50$).
4. **Presence & Frequency Penalties**: Subtracts from a token's logit if it has already appeared in the output, discouraging repetitive loops.

Verified Worked Example: a Tool Loop with Budgets, Error Recovery and Loop Detection

The agent loop above is implemented, run and tested in `examples/ex01_tool_loop.py`. The scripted model first asks for 30 days of history, gets `ERROR: days must be between 1 and 7` back **as an observation**, corrects itself, and answers. Three properties make it safe: a hard `max_steps` budget, errors fed back instead of raised, and a `seen_calls` check that stops identical repeated calls. Tests: `test_tool_loop_recovers_from_tool_error`, `test_tool_loop_step_budget_stops_runaway_agent`.

Pinned for the verified examples: `langgraph=1.2.13`, `mcp=2.3.0`, `pytest=9.1.1` (see `examples/requirements.txt`). All examples run offline with a scripted fake model: `cd examples && pip install -r requirements.txt && pytest -q`.

Reliability arithmetic. With per-step success p , an n -step task succeeds with p^n : $0.95^{10} = 0.60$, $0.99^{10} = 0.90$. This is why the examples spend effort on raising per-step success (clear tool descriptions, helpful errors) and cutting steps.

Further Reading

- [Attention Is All You Need](#)
 - [Hugging Face LLM course](#)
 - [Andrej Karpathy: nanoGPT](#)
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Check Yourself

Answer in your head or on paper first, then open each answer.

▼ 1. What is a token and why does it matter for cost and limits?

A token is a sub-word unit the model reads and writes; pricing, context limits and latency are all counted in tokens.

▼ 2. Why does decoding generate one token at a time?

Each token is sampled conditioned on all previous tokens (autoregression), so generation is sequential even though prompt processing is parallel.

▼ 3. What does temperature do?

It rescales the logits before sampling: low values make output more deterministic, high values more varied.

▼ 4. What is the KV cache?

Stored key and value tensors of earlier tokens so attention does not recompute them each step; it trades memory for speed.